

Chapter 5 Floors: 2023 FBC - Residential, 8th edition | UpCodes

Chapter 5 Floors

Section R501 General

R501.1 Application

The provisions of this chapter shall control the design and construction of the floors for buildings, including the floors of *attic* spaces used to house mechanical or plumbing fixtures and *equipment*.

Exception: Buildings and structures located within the High-Velocity Hurricane Zone shall comply with the provisions of Chapter 44.

R501.2 Requirements

Floor construction shall be capable of accommodating all loads in accordance with Section R301 and of transmitting the resulting loads to the supporting structural elements.

Section R502 Wood Floor Framing

R502.1 General

Wood and wood-based products used for load-supporting purposes shall conform to the applicable provisions of this section.

R502.1.1 Sawn Lumber

Sawn lumber shall be identified by a grade *mark* of an accredited lumber grading or inspection agency and have design values certified by an accreditation body that complies with DOC PS 20. In lieu of a grade *mark*, a certificate of inspection issued by a lumber grading or inspection agency meeting the requirements of this section shall be accepted.

R502.1.1.1 Preservative-Treated Lumber

Preservative treated dimension lumber shall also be identified as required by Section R317.2.

R502.1.1.2 End-Jointed Lumber

Approved end-jointed lumber identified by a grade *mark* conforming to Section R502.1.1 shall be permitted to be used interchangeably with solid-sawn members of the same species and grade. End-jointed lumber used in an assembly required elsewhere in this code to have a fire-resistance rating shall have the designation "Heat-Resistant Adhesive" or "HRA" included in its grade mark.

R502.1.2 Prefabricated Wood I-Joists

Structural capacities and design provisions for prefabricated wood I-joists shall be established and monitored in accordance with ASTM D5055.

R502.1.3 Structural Glued Laminated Timbers

Glued laminated timbers shall be manufactured and identified as required in ANSI A190.1, ANSI 117 and ASTM D3737.

R502.1.4 Structural Log Members

Structural log members shall comply with the provisions of ICC-400.

R502.1.5 Structural Composite Lumber

Structural capacities for structural composite lumber shall be established and monitored in accordance with ASTM D5456.

R502.1.6 Cross-Laminated Timber

Cross-laminated timber shall be manufactured and identified as required by ANSI/APA PRG 320.

R502.1.7 Engineered Wood Rim Board

Engineered wood rim boards shall conform to ANSI/APA PRR 410 or shall be evaluated in accordance with ASTM D7672. Structural capacities shall be in accordance with ANSI/APA PRR 410 or established in accordance with ASTM D7672. Rim boards conforming to ANSI/APA PRR 410 shall be marked in accordance with that standard.

R502.2 Design and Construction

Floors shall be designed in accordance with Sections R301.2.1.1 or ANSI AWC NDS.

Figure R502.2 Floor Construction. Reserved.

R502.2.1 Framing at Braced Wall Lines

Reserved.

R502.2.2 Blocking and Subflooring

Reserved.

R502.3 Allowable Joist Spans

Reserved.

R502.3.1 Sleeping Areas and Attic Joists

Reserved.

Table R502.3.1(1) Floor Joist Spans for Common Lumber Species. Reserved.

Table R502.3.1(2) Floor Joist Spans for Common Lumber Species. Reserved.

R502.3.2 Other Floor Joists

Reserved.

R502.3.3 Floor Cantilevers

Reserved.

Table R502.3.3(1) Cantilever Spans for Floor Joists Supporting Light-Frame Exterior Bearing Wall and Roof Only. Reserved.

Table R502.3.3(2) Cantilever Spans for Floor Joists Supporting Exterior Balcony. Reserved.

R502.4 Joists Under Bearing Partitions

Reserved.

R502.5 Allowable Girder and Header Spans

Reserved.

R502.6 Bearing

Reserved.

R502.6.1 Floor Systems

Reserved.

R502.6.2 Joist Framing

Reserved.

R502.7 Lateral Restraint at Supports

Reserved.

R502.7.1 Bridging

Reserved.

R502.8 Cutting, Drilling and Notching

Reserved.

Figure R502.8 Cutting, Notching and Drilling. Reserved.

R502.8.1 Sawn Lumber

Reserved.

R502.8.2 Engineered Wood Products

Reserved.

R502.9 Fastening

Reserved.

R502.10 Framing of Openings

Reserved.

R502.11 Wood Trusses

R502.11.1 Design

Wood trusses shall be designed in accordance with *approved* engineering practice. The design and manufacture of metal-plate-connected wood trusses shall comply with ANSI/TPI 1. The truss design drawings shall be prepared by a registered professional where required by the *Florida Statutes*.

R502.11.2 Bracing

Trusses shall be braced to prevent rotation and provide lateral stability in accordance with the requirements specified in the *construction documents* for the building and on the individual truss design drawings. In the absence of specific bracing requirements, trusses shall be braced in accordance with accepted industry practices, such as the SBCA *Building Component Safety Information (BCSI) Guide to Good Practice for Handling, Installing & Bracing of Metal Plate Connected Wood Trusses*.

R502.11.3 Alterations to Trusses

Truss members and components shall not be cut, notched, spliced or otherwise altered in any way without the approval of a registered *design professional*. *Alterations* resulting in the addition of load that exceed the design load for the truss, shall not be permitted without verification that the truss is capable of supporting the additional loading.

R502.11.4 Truss Design Drawings

Truss design drawings, prepared in compliance with Section R502.11.1, shall be submitted to the *building official* and *approved* prior to installation. Truss design drawings shall be provided with the shipment of trusses delivered to the job site. Truss design drawings shall include, at a minimum, the information specified as follows:

1. Slope or depth, span and spacing.

2. Location of all joints.
3. Required bearing widths.
4. Design loads as applicable:
 - 4.1. Top chord *live load*.
 - 4.2. Top chord dead load.
 - 4.3. Bottom chord *live load*.
 - 4.4. Bottom chord dead load.
 - 4.5. Concentrated loads and their points of application.
 - 4.6. Controlling wind loads.
5. Adjustments to lumber and joint connector design values for conditions of use.
6. Each reaction force and direction.
7. Joint connector type and description, such as size, thickness or gage, and the dimensioned location of each joint connector except where symmetrically located relative to the joint interface.
8. Lumber size, species and grade for each member.
9. Connection requirements for:
 - 9.1. Truss-to-girder-truss.
 - 9.2. Truss ply-to-ply.
 - 9.3. Field splices.
10. Calculated deflection ratio and/or maximum description for live and total load.
11. Maximum axial compression forces in the truss members to enable the building designer to design the size, connections and anchorage of the permanent continuous lateral bracing. Forces shall be shown on the truss drawing or on supplemental documents.
12. Required permanent truss member bracing location.

R502.12 Draftstopping Required

Draftstopping shall be provided in accordance with Section R302.12.

R502.13 Fireblocking Required

Fireblocking shall be provided in accordance with Section R302.11.

Section R503 Floor Sheathing

R503.1 Design and Construction

Floor sheathing shall be in accordance with Section R301.2.1.1 or ANSI AWC NDS.

Table R503.1 Minimum Thickness of Lumber Floor Sheathing. Reserved.

R503.2 Wood Structural Panel Sheathing

Reserved.

R503.2.1 Identification and Grade

Reserved.

R503.2.1.1 Subfloor and Combined Subfloor Underlayment

Reserved.

Table R503.2.1.1(1) Allowable Spans and Loads for Wood Structural Panels for Roof and Subfloor Sheathing and Combination Subfloor Underlayment. Reserved.

Table R503.2.1.1(2) Allowable Spans for Sanded Plywood Combination Subfloor Underlayment. Reserved.

R503.2.2 Allowable Spans

Reserved.

R503.2.3 Installation

Reserved.

R503.3 Particleboard

Reserved.

R503.3.1 Identification and Grade

Reserved.

R503.3.2 Floor Underlayment

Reserved.

R503.3.3 Installation

Reserved.

Section R504 Pressure Preservative-Treated Wood Floors (On Ground)

R504.1 General

Pressure preservative treated-wood *basement* floors and floors on ground shall be designed to withstand axial forces and bending moments resulting from lateral soil pressures at the base of the exterior walls and floor live and dead loads. Floor framing shall be designed to meet joist deflection requirements in accordance with Section R301.

R504.1.1 Unbalanced Soil Loads

Unless special provision is made to resist sliding caused by unbalanced lateral soil loads, wood *basement* floors shall be limited to applications where the differential depth of fill on opposite exterior foundation walls is 2 feet (610 mm) or less.

R504.1.2 Construction

Joists in wood *basement* floors shall bear tightly against the narrow face of studs in the foundation wall or directly against a band joist that bears on the studs. Plywood subfloor shall be continuous over lapped joists or over butt joints between in-line joists. Sufficient blocking shall be provided between joists to transfer lateral forces at the base of the end walls into the floor system.

R504.1.3 Uplift and Buckling

Where required, resistance to uplift or restraint against buckling shall be provided by interior bearing walls or properly designed stub walls anchored in the supporting soil below.

R504.2 Site Preparation

The area within the foundation walls shall have all vegetation, topsoil and foreign material removed, and any fill material that is added shall be free of vegetation and foreign material. The fill shall be compacted to ensure uniform support of the pressure preservative-treated wood floor sleepers.

R504.2.1 Base

A minimum 4-inch-thick (102 mm) granular base of gravel having a maximum size of $\frac{3}{4}$ inch (19.1 mm) or crushed stone having a maximum size of $\frac{1}{2}$ inch (12.7 mm) shall be placed over the compacted earth.

R504.2.2 Moisture Barrier

Polyethylene sheeting of minimum 6-mil (0.15 mm) thickness shall be placed over the granular base. Joints shall be lapped 6 inches (152 mm) and left unsealed. The polyethylene membrane shall be placed over the pressure preservative-treated wood sleepers and shall not extend beneath the footing plates of the exterior walls.

R504.3 Materials

Framing materials, including sleepers, joists, blocking and plywood subflooring, shall be pressure-preservative treated and dried after treatment in accordance with AWPA U1 (Commodity Specification A, Special Requirement 4.2), and shall bear the *label* of an accredited agency.

Section R505 Cold-Formed Steel Floor Framing

R505.1 General

In accordance with Section R301.2.1.1, the design of cold-formed steel floor framing shall be in accordance with AISI S230, *Standard for Cold-Formed Steel Framing—Prescriptive Method For One- and Two-Family Dwellings*.

Section R506 Concrete Floors (On Ground)

R506.1 General

Concrete slab-on-ground floors shall be designed and constructed in accordance with the provisions of ACI 332 and this section. Floors shall be a minimum $3\frac{1}{2}$ inches (89 mm) thick (for expansive soils, see Section R403.1.8). The specified compressive strength of concrete shall be as set forth in Section R402.2. Footings for concrete slab-on-grade floors shall be in accordance with Chapter 4.

R506.2 Site Preparation

The area within the foundation walls shall have all vegetation, top soil and foreign material removed.

R506.2.1 Fill

Fill material shall be free of vegetation and foreign material. The fill shall be compacted to ensure uniform support of the slab, and except where *approved*, the fill depths shall not exceed 24 inches (610 mm) for clean sand or gravel and 8 inches (203 mm) for earth.

R506.2.2 Base

A 4-inch-thick (102 mm) base course consisting of clean graded sand, gravel, crushed stone, crushed concrete or crushed blast-furnace slag passing a 2-inch (51 mm) sieve shall be placed on the prepared subgrade where the slab is below *grade*.

Exception: A base course is not required where the concrete slab is installed on well-drained or sand-gravel mixture soils classified as Group I according to the United Soil Classification System in accordance with Table R405.1.

R506.2.3 Vapor Retarder

A 6-mil (0.006 inch; 152 µm) polyethylene or *approved* vapor retarder with joints lapped not less than 6 inches (152 mm) shall be placed between the concrete floor slab and the base course or the prepared subgrade where no base course exists.

Exception: The vapor retarder is not required for the following:

1. Garages, utility buildings and other unheated *accessory structures*.
2. For unheated storage rooms having an area of less than 70 square feet (6.5 m²) and carports.
3. Driveways, walks, patios and other flatwork not likely to be enclosed and heated at a later date.
4. Where *approved* by the *building official*, based on local site conditions.

R506.2.4 Joints

R506.2.4.1 Construction Joints

Formed construction joints shall be provided when concrete placing operations are interrupted long enough for previously placed concrete to set.

R506.2.4.2 Contraction Joints

Concrete slabs-on-ground shall be provided with contraction joints in accordance with ACI 332 or other approved industry standards.

Contraction joints are not required for slabs onground provided with crack containment measures complying with one of the following:

1. Concrete slabs on ground shall contain micro- or macro-synthetic fiber reinforcement. Fiber lengths shall be $\frac{1}{2}$ inch to 2.25 inches (13 to 56 mm) in length. Dosage amounts shall be from 0.75 to 3.0 pounds per cubic yard (0.45 to 1.78 kg/m³) in accordance with the manufacturer's recommendations. Synthetic fibers shall comply with ASTM C1116. The manufacturer or supplier shall provide certification of compliance with ASTM C1116 when requested by the building official; or,
2. Concrete slabs on ground containing 6 × 6 W1.4 × W1.4 welded wire reinforcement fabric located in the middle to the upper $\frac{1}{3}$ of the slab. Welded wire reinforcement fabric shall be supported with approved materials or supports at spacing of 3 feet (914 mm) or less. Welded plain wire reinforcement fabric for concrete shall conform to ASTM A1064/A1064 M Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.

Exception: Where random cracking is acceptable to the building owner, contraction joints are not required where crack containment complying with one of the above methods is provided.

Section R507 Exterior Decks

R507.1 Decks

Wood-framed decks shall be in accordance with this section. For decks using materials and conditions not prescribed in this section, refer to Section R301.

R507.2 Plastic Composite Deck Boards, Stair Treads, Guards, or Handrails

Plastic composite exterior deck boards, stair treads, guards and handrails shall comply with the requirements of ASTM D7032 and the requirements of this section.

R507.2.1 Labeling

Plastic composite deck boards and stair treads, or their packaging, shall bear a label that indicates compliance to ASTM D7032 and includes the allowable load and maximum allowable span determined in accordance with ASTM D7032. Plastic or composite handrails and guards, or their packaging, shall bear a label that indicates compliance to ASTM D7032 and includes the maximum allowable span determined in accordance with ASTM D7032.

R507.2.2 Flame Spread Index

Plastic composite deck boards, stair treads, guards and handrails shall exhibit a flame spread index not exceeding 200 when tested in accordance with ASTM E84 or UL 723 with the test specimen remaining in place during the test.

Exception: Plastic composites determined to be noncombustible.

R507.2.3 Decay Resistance

Plastic composite deck boards, stair treads, guards and handrails containing wood, cellulosic or other biodegradable materials shall be decay resistant in accordance with ASTM D7032.

R507.2.4 Termite Resistance

Where required by Section R318, plastic composite deck boards, stair treads, guards and handrails containing wood, cellulosic or other biodegradable materials shall be termite resistant in accordance with ASTM D7032.

R507.2.5 Installation of Plastic Composites

Plastic composite deck boards, stair treads, guards and handrails shall be installed in accordance with this code and the manufacturer's instructions.

R507.3 Deck Footings

Deck footings shall be sized to carry the imposed loads from the deck structure to the ground as shown in Figure R507.3. The footing depth shall be in accordance with Section R403.1.4.

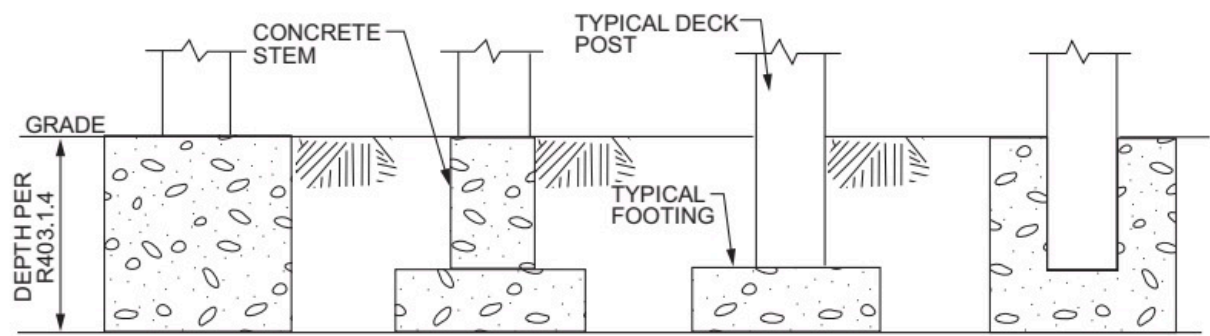


FIGURE R507.3

TYPICAL DECK POSTS TO DECK FOOTINGS

R507.4 Deck Posts

For single-level wood-framed decks with beams sized in accordance with Table R507.5, deck post size shall be in accordance with Table R507.4.

TABLE R507.4
DECK POST HEIGHT^a

DECK POST SIZE	MAXIMUM HEIGHT ^a
4 × 4	8'
4 × 6	8'
6 × 6	14'

For SI: 1 foot = 304.8 mm.

a. Measured to the underside of the beam.

R507.4.1 Deck Post to Deck Footing Connection

Posts shall bear on footings in accordance with Section R403 and Figure R507.3. Posts shall be restrained to prevent lateral displacement at the bottom support. Such lateral restraint shall be provided by manufactured connectors installed in accordance with Section R507 and the manufacturers' instructions or a minimum post embedment of 12 inches (305 mm) in surrounding soils or concrete piers.

R507.5 Deck Beams

Maximum allowable spans for wood deck beams, as shown in Figure R507.5, shall be in accordance with Table R507.5. Beam plies shall be fastened with two rows of 10d (3-inch × 0.128-inch) nails minimum at 16 inches (406 mm) on center along each edge. Beams shall be permitted to cantilever at each end up to one-fourth of the actual beam span. Splices of multispan beams shall be located at interior post locations.

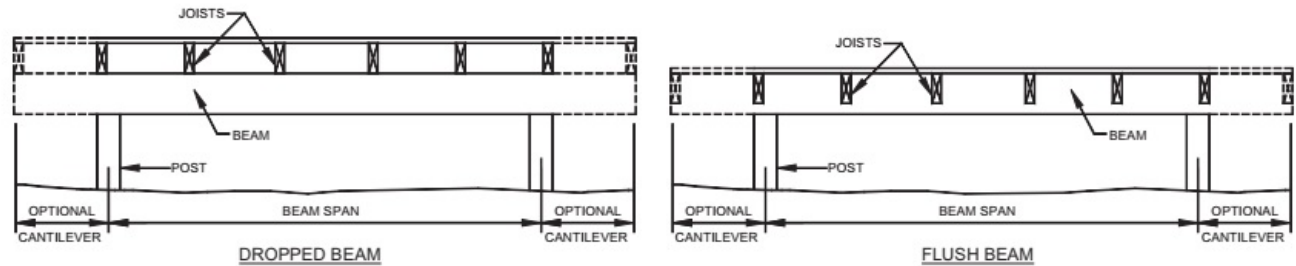


FIGURE R507.5

TYPICAL DECK BEAM SPANS

TABLE R507.5
DECK BEAM SPAN LENGTHS

SPECIES ^c	SIZE ^d	EFFECTIVE DECK JOIST SPAN LENGTH ^a
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		6	8	10	12	14	16	18
		DECK BEAM SPAN LENGTH ^{a, b} (feet-inches)						
Southern pine	2 — 2 × 6	6-11	5-11	5-4	4-10	4-6	4-3	4-0
	2 — 2 × 8	8-9	7-7	6-9	6-2	5-9	5-4	5-0
	2 — 2 × 10	10-4	9-0	8-0	7-4	6-9	6-4	6-0
	2 — 2 × 12	12-2	10-7	9-5	8-7	8-0	7-6	7-0
	3 — 2 × 6	8-2	7-5	6-8	6-1	5-8	5-3	5-0
	3 — 2 × 8	10-10	9-6	8-6	7-9	7-2	6-8	6-4
	3 — 2 × 10	13-0	11-3	10-0	9-2	8-6	7-11	7-6
	3 — 2 × 12	15-3	13-3	11-10	10-9	10-0	9-4	8-10
Douglas fir-larch ^e hem-fir ^e Spruce-pine-fir ^e redwood, western cedars,ponderosa pine ^f red pine ^f	3 × 6 or 2 — 2 × 6	5-5	4-8	4-2	3-10	3-6	3-1	2-9
	3 × 8 or 2 — 2 × 8	6-10	5-11	5-4	4-10	4-10	4-1	3-8
	3 × 10 or 2 — 2 × 10	8-4	7-3	6-6	5-11	5-6	5-1	4-8
	3 × 12 or 2 — 2 × 12	9-8	8-5	7-6	6-10	6-4	5-11	5-7
	4 × 6	6-5	5-6	4-11	4-6	4-2	3-11	3-8
	4 × 8	8-5	7-3	6-6	5-11	5-6	5-2	4-10
	4 × 10	9-11	8-7	7-8	7-0	6-6	6-1	5-8
	4 × 12	11-5	9-11	8-10	8-1	7-6	7-0	6-7
	3 — 2 × 6	7-4	6-8	6-0	5-6	5-1	4-9	4-6
	3 — 2 × 8	9-8	8-6	7-7	6-11	6-5	6-0	5-8
	3 — 2 × 10	12-0	10-5	9-4	8-6	7-10	7-4	6-11
	3 — 2 × 12	13-11	12-1	10-9	9-10	9-1	8-6	8-1

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, 1 pound = 0.454 kg.

- Live load = 40 psf, dead load = 10 psf, $L/\Delta = 360$ at main span, $L/\Delta = 180$ at cantilever with a 220-pound point load applied at the end.
- Beams supporting deck joists from one side only.
- No. 2 grade, wet service factor.
- Beam depth shall be greater than or equal to depth of joists with a flush beam condition.
- Includes incising factor.
- Northern species. Incising factor not included.
- For calculation of effective deck joist span, the actual joist span length shall be multiplied by the joist span factor from the following table.

C/J	JOIST SPAN FACTOR
0 (no cantilever)	0.66
1/12 (0.87)	0.72
1/10 (0.10)	0.80
1/8 (0.125)	0.84
1/6 (0.167)	0.90
1/4 (0.250)	1.00

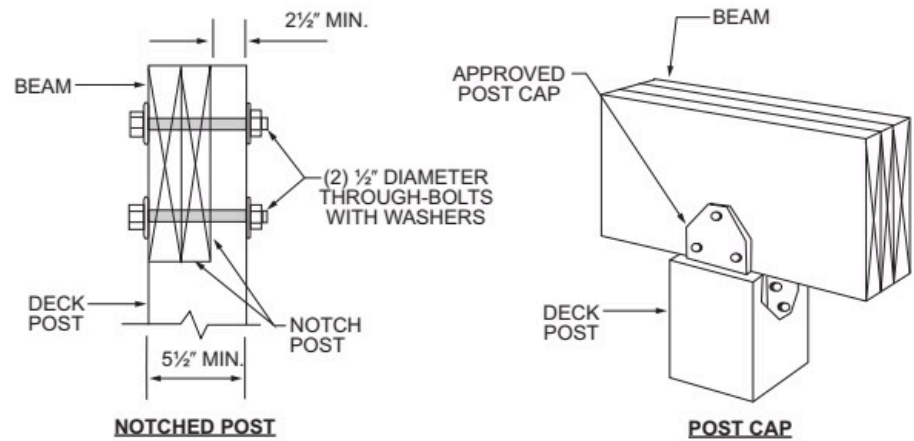
J = actual joist span length (feet)

C = actual joist cantilever length (feet)

R507.5.1 Deck Post to Deck Beam

Deck beams shall be attached to deck posts in accordance with Figure R507.5.1 or by other equivalent means capable to resist lateral displacement. Manufactured post-to-beam connectors shall be sized for the post and beam sizes. All bolts shall have washers under the head and nut.

Exception: Where deck beams bear directly on footings in accordance with Section R507.4.1.



For SI: 1 inch = 25.4 mm.

FIGURE R507.5.1
DECK BEAM TO DECK POST

R507.6 Deck Joists

Maximum allowable spans for wood deck joists, as shown in Figure R507.6, shall be in accordance with Table R507.6. Maximum allowable spacing for joists shall be limited by the decking material in accordance with Table R507.7. Deck joists shall be permitted to cantilever not greater than one-fourth of the actual, adjacent joist span.

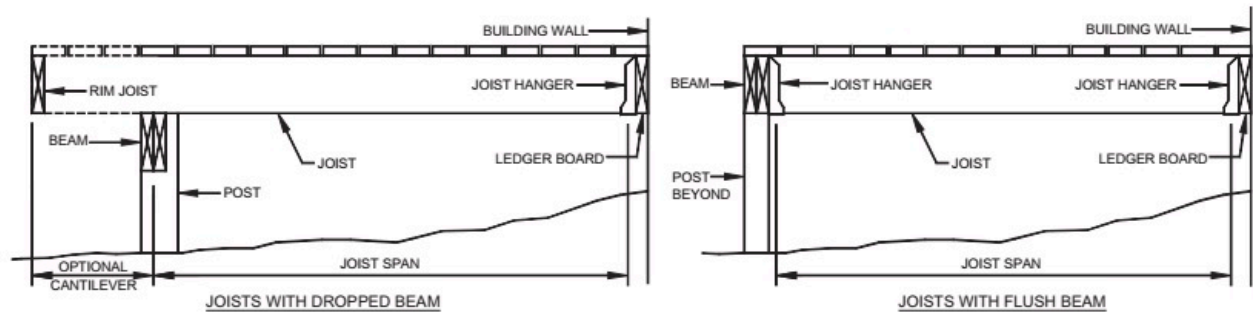


FIGURE R507.6
TYPICAL DECK JOIST SPANS
TABLE R507.6
DECK JOIST SPANS FOR COMMON LUMBER SPECIES (ft. - in.)

SPECIES ^a	SIZE	SPACING OF DECK JOISTS WITH NO CANTILEVER ^{b,f} (inches)			SPACING OF DECK JOISTS WITH CANTILEVERS ^c (inches)		
		12	16	24	12	16	24
Southern pine	2 × 6	9-11	9-0	7-7	6-8	6-8	6-8
	2 × 8	13-1	11-10	9-8	10-1	10-1	9-8
	2 × 10	16-2	14-0	11-5	14-6	14-0	11-5
	2 × 12	18-0	16-6	13-6	18-0	16-6	13-6
Douglas fir-larch, ^d hemfir spruce-pine-fir ^d	2 × 6	9-6	8-8	7-2	6-3	6-3	6-3
	2 × 8	12-6	11-1	9-1	9-5	9-5	9-1

	2 × 10	15-8	13-7	11-1	13-7	13-7	11-1
	2 × 12	18-0	15-9	12-10	18-0	15-9	12-10
Redwood, western cedars, ponderosa pine, ^e red pine ^e	2 × 6	8-10	8-0	7-0	5-7	5-7	5-7
	2 × 8	11-8	10-7	8-8	8-6	8-6	8-6
	2 × 10	14-11	13-0	10-7	12-3	12-3	10-7
	2 × 12	17-5	15-1	12-4	16-5	15-1	12-4

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, 1 pound = 0.454 kg.

- a. No. 2 grade with wet service factor.
- b. Live load = 40 psf, dead load = 10 psf, $L/\Delta = 360$.
- c. Live load = 40 psf, dead load = 10 psf, $L/\Delta = 360$ at main span, $L/\Delta = 180$ at cantilever with a 220-pound point load applied to end.
- d. Includes incising factor.
- e. Northern species with no incising factor.
- f. Cantilevered spans not exceeding the nominal depth of the joist are permitted.

R507.6.1 Lateral Restraint at Supports

Joist ends and bearing locations shall be provided with lateral restraint to prevent rotation. Where lateral restraint is provided by joist hangers or blocking between joists, their depth shall equal not less than 60 percent of the joist depth. Where lateral restraint is provided by rim joists, they shall be secured to the end of each joist with not less than (3) 10d (3-inch × 0.128-inch) nails or (3) No. 10 × 3-inch (76 mm) long wood screws.

R507.6.2 Deck Joist and Deck Beam Bearing

The ends of each joist and beam shall have not less than $1\frac{1}{2}$ inches (38 mm) of bearing on wood or metal and not less than 3 inches (76 mm) on concrete or masonry for the entire width of the beam. Joist framing into the side of a ledger board or beam shall be supported by approved joist hangers. Joists bearing on a beam shall be connected to the beam to resist lateral displacement.

R507.7 Decking

Maximum allowable spacing for joists supporting decking shall be in accordance with Table R507.7. Wood decking shall be attached to each supporting member with not less than (2) 8d threaded nails or (2) No. 8 wood screws.

TABLE R507.7
MAXIMUM JOIST SPACING

MATERIAL TYPE AND NOMINAL SIZE	MAXIMUM ON-CENTER JOIST SPACING	
	Perpendicular to joist	Diagonal to joist ^a
$1\frac{1}{4}$ -inch-thick wood	16 inches	12 inches
2-inch-thick wood	24 inches	16 inches
Plastic composite	In accordance with Section R507.2	In accordance with Section R507.2

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 degree = 0.01745 rad.

- a. Maximum angle of 45 degrees from perpendicular for wood deck boards

R507.8 Vertical and Lateral Supports

Where supported by attachment to an exterior wall, decks shall be positively anchored to the primary structure and designed for both vertical and lateral loads. Such attachment shall not be accomplished by the use of toenails or nails subject to withdrawal. For decks with cantilevered framing members, connection to exterior walls or other framing members shall be designed and constructed to resist uplift resulting from the full live load specified in Table R301.5 acting on the cantilevered portion of the deck. Where positive connection to the primary building structure cannot be verified during inspection, decks shall be self-supporting.

R507.8.1 Vertical Supports

Vertical loads shall be transferred to band joists with ledgers in accordance with this section.

R507.8.1.1 Ledger Details

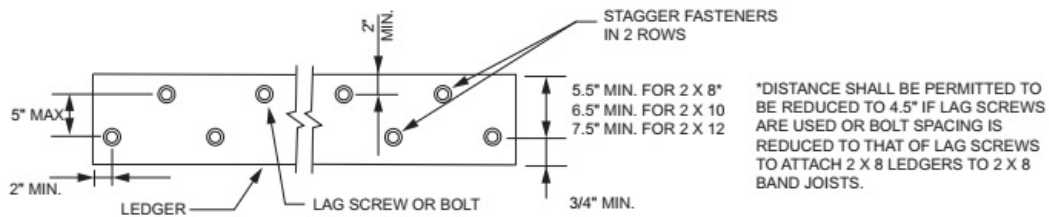
Deck ledgers shall be a minimum 2-inch by 8-inch (51 mm by 203 mm) nominal, pressure-preservative-treated southern pine, incised pressure-preservative-treated Hem-fir, or approved, naturally durable, No. 2 grade or better lumber. Deck ledgers shall not support concentrated loads from beams or girders. Deck ledgers shall not be supported on stone or masonry veneer.

R507.8.1.2 Band Joist Details

Band joists attached by a ledger shall be a minimum 2-inch-nominal (51 mm), solid-sawn, spruce-pine-fir lumber or a minimum 1-inch (25 mm) nominal engineered wood rim boards specified in Section R502.1.7. Band joists attached by a ledger shall be fully supported by a wall or sill plate below.

R507.8.1.3 Ledger to Band Joist Fastener Details

Fasteners used in deck ledger connections in accordance with Table R507.8.1.3(1) shall be hot-dipped galvanized or stainless steel and shall be installed in accordance with Table R507.8.1.3(2) and Figures R507.8.1.3(1) and R507.8.1.3(2).



For SI: 1 inch = 25.4 mm.

FIGURE R507.8.1.3(1)

PLACEMENT OF LAG SCREWS AND BOLTS IN LEDGERS

TABLE R507.8.1.3(1)

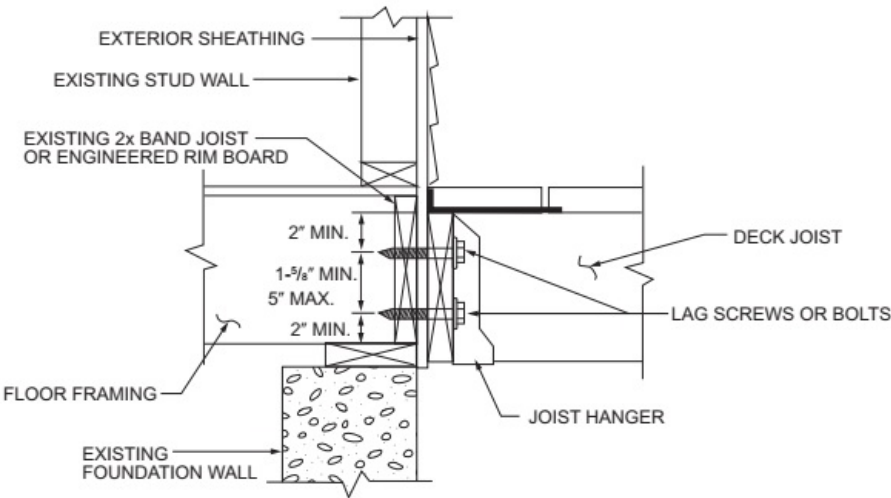
DECK LEDGER CONNECTION TO BAND JOIST^{a, b}

(Deck live load = 40 psf, deck dead load = 10 psf)

CONNECTION DETAILS	JOIST SPAN						
	6' and less	6'1" to 8'	8'1" to 10'	10'1" to 12'	12'1" to 14'	14'1" to 16'	16'1" to 18'
	On-center spacing of fasteners						
1/2-inch diameter lag screw with 1/2-inch maximum sheathing ^{c, d}	30	23	18	15	13	11	10
1/2-inch diameter bolt with 1/2-inch maximum sheathing ^d	36	36	34	29	24	21	19
1/2-inch diameter bolt with 1-inch maximum sheathing ^e	36	36	29	24	21	18	16

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa.

- a. Ledgers shall be flashed in accordance with Section R703.4 to prevent water from contacting the house band joist.
- b. Reserved.
- c. The tip of the lag screw shall fully extend beyond the inside face of the band joist.
- d. Sheathing shall be wood structural panel or solid sawn lumber.
- e. Sheathing shall be permitted to be wood structural panel, gypsum board, fiberboard, lumber or foam sheathing. Up to 1/2-inch thickness of stacked washers shall be permitted to substitute for up to 1/2 inch of allowable sheathing thickness where combined with wood structural panel or lumber sheathing.



For SI: 1 inch = 25.4 mm.

FIGURE R507.8.1.3(2)

PLACEMENT OF LAG SCREWS AND BOLTS IN BAND JOISTS

TABLE R507.8.1.3(2)

PLACEMENT OF LAG SCREWS AND BOLTS IN DECK LEDGERS AND BAND JOISTS

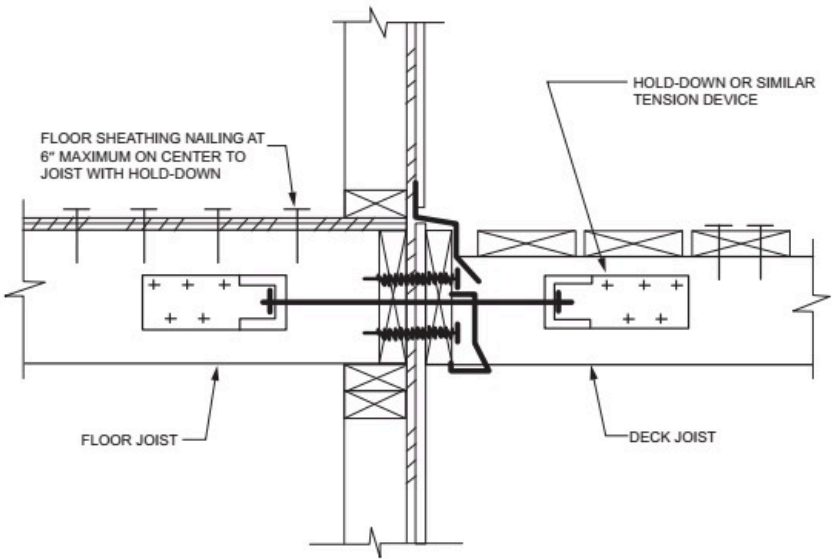
MINIMUM END AND EDGE DISTANCES AND SPACING BETWEEN ROWS				
	TOP EDGE	BOTTOM EDGE	ENDS	ROW SPACING
Ledger ^a	2 inches ^d	³ / ₄ inch	2 inches ^b	1 ⁵ / ₈ inches ^b
Band Joist ^c	³ / ₄ inch	2 inches	2 inches _b	1 ⁵ / ₈ inches ^b

For SI: 1 inch = 25.4 mm

- a. Lag screws or bolts shall be staggered from the top to the bottom along the horizontal run of the deck ledger in accordance with Figure R507.8.1.3(1)
- b. Maximum 5 inches.
- c. For engineered rim joists, the manufacturer's recommendations shall govern.
- d. The minimum distance from bottom row of lag screws or bolts to the top edge of the ledger shall be in accordance with Figure R507.8.1.3(1).

R507.8.2 Deck Lateral Load Connection

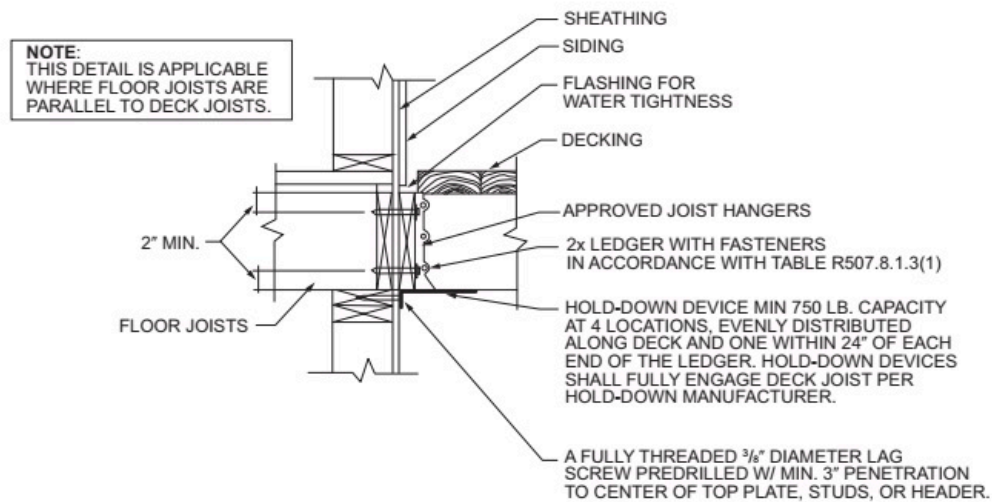
Where the lateral load connection is provided in accordance with Figure R507.8.2(1), hold-down tension devices shall be installed in not less than two locations per deck, within 24 inches of each end of the deck. Each device shall have an allowable stress design capacity of not less than 1,500 pounds (6672 N). Where the lateral load connections are provided in accordance with Figure R507.8.2(2), the hold-down tension devices shall be installed in not less than four locations per deck, and each device shall have an allowable stress design capacity of not less than 750 pounds (3336 N).



For SI: 1 inch = 25.4 mm.

FIGURE 507.8.2(1)

DECK ATTACHMENT FOR LATERAL LOADS



For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

FIGURE R507.8.2(2)
DECK ATTACHMENT FOR LATERAL LOADS

R507.9 Exterior Guards

Guards shall be constructed to meet the requirements of Sections R301.5 and R312, and this section.

R507.9.1 Support of Guards

Where guards are supported on deck framing, guard loads shall be transferred to the deck framing with a continuous load path to the deck joists.

R507.9.1.1 Guards Supported by Side of Deck Framing

Where guards are connected to the interior or exterior side of a deck joist or beam, the joist or beam shall be connected to the adjacent joists to prevent rotation of the joist or beam. Connections relying only on fasteners in end grain withdrawal are not permitted.

R507.9.1.2 Guards Supported on Top of Deck Framing

Where guards are mounted on top of the decking, the guards shall be connected to the deck framing or blocking and installed in accordance with manufacturer's instructions to transfer the guard loads to the adjacent joists.

R507.9.2 Wood Posts at Deck Guards

Where 4-inch by 4-inch (102 mm by 102 mm) wood posts support guard loads applied to the top of the guard, such posts shall not be notched at the connection to the supporting structure.

R507.9.3 Plastic Composite Guards

Plastic composite guards shall comply with the provisions of Section R507.2.

R507.9.4 Other Guards

Other guards shall be in accordance with either manufacturer's instructions or accepted engineering principles.